

Lilly Events Feed — Anomaly Detection Strategy Document

Overview

This document provides a structured anomaly detection strategy for the Lilly Events Feed using AWS Glue Data Quality anomaly detection metrics.

The goal of anomaly detection is to identify unusual, invalid, unexpected, or risky patterns in event and attendee data before they impact downstream reporting, analytics, compliance, operational workflows, or business decision-making.

The Lilly Events Feed contains event lifecycle information, attendee information, product mappings, geography information, event ownership details, and platform metadata. Because of this, anomaly detection plays a critical role in ensuring data quality, operational reliability, and business trust.

1. Available AWS Anomaly Search Metrics

The following anomaly search metrics are officially supported by AWS Glue Data Quality and can be used to monitor, profile, and detect abnormal behavior in enterprise data feeds.

Metric	Description
ColumnLength	Measures the length characteristics of column values and helps detect abnormal text sizes, corrupted payloads, truncated values, or unexpected content growth. Useful for identifying malformed event names, oversized links, or corrupted metadata fields.
ColumnValues	Validates whether column values match expected or allowed values. Useful for detecting invalid event statuses, unsupported attendee statuses, unexpected countries, or corrupted categorical data.
Completeness	Measures the percentage of non-null values in a column. Helps detect missing data spikes, incomplete records, ingestion failures, or upstream ETL issues affecting critical business columns.
Distinct ValuesCount	Measures the number of distinct values present in a column. Useful for identifying sudden cardinality changes, unexpected new categories, or abnormal reductions in business diversity.
RowCount	Measures the total number of records in a dataset or partition. Helps detect feed failures, duplicate ingestion, delayed processing, or abnormal spikes in event data volume.

Metric	Description
Uniqueness	Measures whether column values remain unique across records. Commonly used for identifying duplicate event identifiers, repeated attendee records, or data duplication issues.
Mean	Calculates the average value of numeric columns. Useful for identifying statistical drift or unexpected changes in numerical business metrics over time.
Sum	Calculates the total aggregated value of numeric columns. Helps detect abnormal increases or decreases in total metrics such as attendee counts or event-related aggregations.
StandardDeviation	Measures how much numeric data varies from the average. Useful for identifying unstable distributions, abnormal fluctuations, or sudden variability changes in event behavior.
Entropy	Measures the distribution randomness or diversity of values within a column. Helps detect distribution drift such as unusual country concentration, product imbalance, or abnormal event-type dominance.
Unique ValueRatio	Measures the ratio between unique values and total records. Useful for detecting duplicate spikes, identity reuse, or sudden reductions in data uniqueness.
ColumnCount	Measures the total number of columns in a dataset. Helps identify schema changes, missing columns, newly added columns, or upstream structural modifications.
ColumnCorrelation	Measures relationships between two columns and identifies unexpected behavioral drift between related business entities. Useful for detecting mapping inconsistencies or relationship anomalies.
CustomSql	Enables implementation of business-specific validation rules using SQL logic. Useful for enforcing date validations, workflow rules, lifecycle consistency, and enterprise governance checks.
AllStatistic	Generates a combined statistical profile using multiple metrics simultaneously. Useful for broad anomaly monitoring, automated profiling, and enterprise-level observability.

2. Top 20 Business-Critical Anomalies for Lilly Events Feed

#	Anomaly	AWS Metric
1	Sudden event feed row drop/spike	RowCount

#	Anomaly	AWS Metric
2	Duplicate <code>event_id</code> increase	Uniqueness
3	NULL spike in <code>event_id</code>	Completeness
4	NULL spike in <code>product_id</code>	Completeness
5	NULL spike in <code>event_country</code>	Completeness
6	Invalid <code>event_status</code> values appearing	ColumnValues
7	Invalid <code>attendee_status</code> values appearing	ColumnValues
8	Sudden increase in CANCELLED events	Entropy
9	Unexpected new <code>event_type</code> values	Distinct ValuesCount
10	Unexpected new <code>event_sub_type</code> values	Distinct ValuesCount
11	Country distribution drift	Entropy
12	Product distribution drift	Entropy
13	Duplicate attendee ratio increase	Unique ValueRatio
14	Sudden drop in unique events	Distinct ValuesCount
15	Missing <code>event_owner_id</code> spike	Completeness
16	Abnormal <code>event_name</code> length increase	ColumnLength
17	Event-provider relationship drift	ColumnCorrelation
18	Event-country and attendee-country mismatch drift	ColumnCorrelation
19	Unexpected schema/column changes	ColumnCount
20	Invalid event lifecycle (<code>event_end_date < event_start_date</code>)	CustomSql

3. Detailed Explanation of Each Recommended Anomaly

1. Sudden Event Feed Row Drop/Spike

Metric: RowCount

Detects abnormal increases or decreases in the total number of event records being ingested. This anomaly is critical for identifying feed failures, delayed pipelines, duplicate ingestion, or upstream extraction issues.

2. Duplicate `event_id` Increase

Metric: Uniqueness

Detects situations where event identifiers that are expected to remain unique start appearing multiple times. This helps prevent duplicate reporting, incorrect aggregation, and downstream analytics corruption.

3. NULL Spike in `event_id`

Metric: Completeness

Monitors missing values in the primary event identifier column. Missing event identifiers can break event traceability, reporting accuracy, and downstream joins.

4. NULL Spike in `product_id`

Metric: Completeness

Detects missing product mappings associated with events. Product attribution is critical for business analytics, regulatory tracking, and commercial reporting.

5. NULL Spike in `event_country`

Metric: Completeness

Identifies missing country information in event records. Country-level reporting is essential for regional analysis, compliance monitoring, and operational visibility.

6. Invalid `event_status` Values Appearing

Metric: ColumnValues

Detects unexpected or unsupported event statuses entering the feed. This helps identify upstream source corruption, schema changes, or workflow inconsistencies.

7. Invalid `attendee_status` Values Appearing

Metric: ColumnValues

Monitors attendee workflow states for unexpected values. This helps prevent reporting inconsistencies and broken attendee lifecycle tracking.

8. Sudden Increase in CANCELLED Events

Metric: Entropy

Detects abnormal spikes in cancelled events compared to historical behavior. This may indicate operational disruptions, business issues, or upstream workflow failures.

9. Unexpected New `event_type` Values

Metric: Distinct ValuesCount

Detects new event types appearing in the feed that were not historically present. This helps identify uncontrolled source changes or newly introduced business categories.

10. Unexpected New `event_sub_type` Values

Metric: Distinct ValuesCount

Monitors sudden increases in event subtype diversity. Useful for identifying schema drift, taxonomy expansion, or source-system inconsistencies.

11. Country Distribution Drift

Metric: Entropy

Detects abnormal shifts in the distribution of events across countries. This helps identify regional ingestion issues, unexpected concentration patterns, or missing country data.

12. Product Distribution Drift

Metric: Entropy

Monitors abnormal changes in product-level event distributions. Useful for detecting mapping corruption, source inconsistencies, or abnormal commercial activity.

13. Duplicate Attendee Ratio Increase

Metric: Unique ValueRatio

Detects rising duplication among attendee records within events. This helps identify registration duplication, ingestion retries, or identity reuse problems.

14. Sudden Drop in Unique Events

Metric: Distinct ValuesCount

Detects situations where the number of distinct events drops unexpectedly. This often indicates missing data partitions, partial ingestion, or pipeline failures.

15. Missing `event_owner_id` Spike

Metric: Completeness

Monitors ownership metadata completeness. Missing ownership information affects governance, accountability, and operational management.

16. Abnormal `event_name` Length Increase

Metric: ColumnLength

Detects unexpected growth in event name size. This may indicate payload corruption, HTML injection, malformed source data, or ingestion formatting issues.

17. Event-Provider Relationship Drift

Metric: ColumnCorrelation

Monitors changing relationships between events and providers. Unexpected correlation changes may indicate mapping corruption or incorrect provider assignments.

18. Event-Country and Attendee-Country Mismatch Drift

Metric: ColumnCorrelation

Detects abnormal changes in the relationship between event geography and attendee geography. Useful for identifying incorrect mappings, ingestion issues, or unusual participation patterns.

19. Unexpected Schema/Column Changes

Metric: ColumnCount

Detects sudden changes in the dataset structure such as missing columns or newly introduced fields. Schema anomalies can break downstream pipelines and analytics processes.

20. Invalid Event Lifecycle

`( event_end_date < event_start_date )`

Metric: CustomSql

Uses custom SQL validation logic to detect invalid event date ranges. This anomaly helps maintain lifecycle consistency and prevents invalid scheduling data from entering downstream systems.

Conclusion

The recommended anomaly detection strategy for the Lilly Events Feed focuses on business-critical data quality monitoring using AWS Glue Data Quality metrics.

The implementation should prioritize:

1. Feed health monitoring
2. Mandatory field completeness
3. Duplicate prevention
4. Business-rule validation
5. Distribution drift detection
6. Relationship consistency
7. Schema stability

By implementing these anomaly checks, the organization can significantly improve data reliability, operational visibility, governance, and downstream analytical trust across enterprise event systems.